

Annie Cai

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Education

University of Waterloo

Waterloo, ON

B.ASc in Systems Design Engineering - GPA: 3.99/4.00, 94%

2022 - 2027

- *Awards:* 4x Term Dean's List, President's Undergraduate Research Award, Faculty of Engineering Scholarship, President's Distinction Scholarship
- *Relevant Courses:* Data Structures, Algorithms, Databases, OS, Networks, Distributed Systems

Skills

Languages: Python, Go, C++, Java, TypeScript, JavaScript, SQL, GraphQL, HTML/CSS

Technologies: Apache Spark, Apache Kafka, React, Flask, Node.js, Next.js, Express.js, Django

Tools: Git, AWS, PostgreSQL, MySQL, MongoDB, Kubernetes, Docker, Redis, Envoy, Jenkins, Terraform

Work Experience

Snowflake

Bellevue, WA

Software Engineer Intern

Sep. 2025 - Dec. 2025

- Engineering scalable, low-latency load testing pipelines to stress-test Envoy ingress load balancing and strengthen global traffic resilience across multi-cloud environments (Go, Python, C++)

Datadog

New York, NY

Software Engineer Intern

Jan. 2025 - Apr. 2025

- Architected a **Kafka** streaming service, **processing 100k+ messages/sec** and achieving millisecond-level latency across distributed systems
- Implemented **Go-level concurrency** in the processing pipeline, leading to a **2x CPU utilization drop**
- Spearheaded the development of a **distributed, multilayer caching system**, using hashing and LRU eviction to boost cache hit rate and reduce data lookup latency
- Leveraged Kubernetes clusters and Docker containers to deploy and orchestrate the service

Newfront (YC W18)

San Francisco, CA

Software Engineer Intern

May. 2024 - Aug. 2024

- Built a real-time messaging system using **Websockets**, with scalable pub/sub message routing, to allow bidirectional communication for an AI chatbot
- Refactored **Python** API endpoints, enabling concurrent execution, **reducing response times by 3x**
- Improved **MySQL** database read/write speeds through strategic indexing and schema optimizations

Royal Bank of Canada

Toronto, ON

Software Engineer Intern

Sep. 2023 - Dec. 2023

- Streamlined ETL pipeline by modularizing the **Python** pipeline, enabling parallel and independent execution and enhancing workflow efficiency
- Achieved a **40% cost reduction** in storage expenses by **migrating 8GB+** of data to **AWS S3 buckets**
- Scaled database retrieval job during high load periods using **Redis** and **Apache Spark**, leveraging resource tuning and caching strategies to **lower job time from hours to under 5 minutes**

Extracurriculars

UW Blueprint React, TypeScript, Express.js, PostgreSQL, Docker, Heroku

Directed a cross-functional student team to ship a full-stack pet management platform for a nonprofit.

Undergraduate Researcher C++

Operating systems research, focusing on synchronization techniques and locks for persistent memory.